

VISVESVARAYA TECHNOLOGICAL UNIVERSITY, BELAGAVI

VTU Textbook Series | Computer Science & Engineering

MODULE 1

ARTIFICIAL INTELLIGENCE

Introduction to AI and Applications

Beyond Classical AI — Machine Intelligence, Knowledge
Representation & Search Algorithms

PART A · Introduction to AI

PART B · Machine Intelligence

PART C · Knowledge Representation

B.E. / B.Tech Semester I / II • Course Code 1BAIA103 / 203
2025-2026 Scheme

Bloom's Taxonomy L1-L6

GATE / Placement Questions

Previous Year VTU Questions

Dr. Lokesh M R

Professor, Dept. of Information Science & Engineering
A J Institute of Engineering and Technology, Mangaluru

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Introduction to AI and Applications — Module 1

Beyond Classical AI: Machine Intelligence, Knowledge Representation & Search Algorithms

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Visvesvaraya Technological University Textbook Series | Computer Science & Engineering
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AI · Machine Intelligence · Knowledge Representation · Search Algorithms



Dedicated to
every curious student who dares to ask
“Can machines think?”

Preface

Artificial Intelligence (AI) has moved from science fiction into the fabric of everyday life. From voice assistants and self-driving vehicles to medical diagnostics and intelligent search engines, AI shapes the decisions, products and services that define the modern world. Yet understanding the foundational principles behind these technologies remains a challenge for many students entering the field.

This textbook, Introduction to AI and Applications (Module 1), is designed as Module 1 of a comprehensive undergraduate course in Artificial Intelligence. It presents three carefully structured parts:

- **Part A — Introduction to Artificial Intelligence** — origins, definitions, history, types, and the interplay between AI, augmented intelligence and machine learning.
- **Part B — Machine Intelligence** — the nature of intelligence, rational agents, search strategies, and both uninformed and informed search algorithms.
- **Part C — Knowledge Representation** — how AI agents store, reason with and apply knowledge to act intelligently in the real world.

A distinctive feature of this formatted edition is the integration, after every topic, of three practice blocks: Bloom's Taxonomy questions at all six cognitive levels (L1 recall to L6 creation); GATE / placement-practice questions mapped to those same Bloom levels; and previous-year VTU questions. Together they support classroom discussion, internal assessment, GATE and placement preparation, and independent study.

The content is aligned with the VTU 2025-2026 scheme for B.E./B.Tech in Computer Science and related disciplines. Each concept is explained from first principles and supported by numbered tables, embedded figures, worked examples and real-world case studies that bridge theory and practice.

The author hopes this book serves not only as a trusted academic companion but also as a launching pad for students who wish to explore AI research, build intelligent systems, or develop a deeper appreciation for the technology reshaping our world.

Dr. Lokesh M R

Mangaluru, Karnataka — 2026

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How to Use This Edition

This edition reorganises Module 1 into three parts. After every topic you will find three practice blocks:

- **Bloom's Taxonomy questions** — span six cognitive levels — Remember (L1), Understand (L2), Apply (L3), Analyse (L4), Evaluate (L5), Create (L6) — presented as compact numbered lists.
- **GATE / placement questions** — are CS/IT and Data-Aptitude style MCQs, numericals and NATs, each tagged with a Bloom level, with the answer/hint following the question in italics.
- **Previous-year VTU questions** — show the wording, weightage and expectation of end-semester and internal-examination questions, as compact bullet lists.

To keep the book concise, all three question blocks use the same compact list style (instead of tables), and text-heavy source figures (timelines, box-and-arrow diagrams) have been converted to selectable text, tables and redrawn text diagrams. The genuine node-and-edge search trees (DFS, DLS, BFS, UCS) are retained as figures.

The following legend is used throughout the book:

Table 0.1 Bloom's Taxonomy Legend Used in This Edition

Bloom Level	Cognitive Process	Typical Question Signal
L1	Remember (Recall)	Define, list, name, recall
L2	Understand	Explain, describe, classify, summarise
L3	Apply	Apply, solve, use, demonstrate
L4	Analyse	Compare, differentiate, categorise, examine
L5	Evaluate	Judge, justify, assess, critique
L6	Create	Design, construct, develop, propose

PART A

Introduction to Artificial Intelligence

Origins, definitions, history, types of AI, and how AI relates to augmented intelligence, cognitive computing, machine learning and deep learning.

CHAPTER 1

Introduction to Artificial Intelligence

1.1 What is Artificial Intelligence?

In 2004, John McCarthy defined Artificial Intelligence (AI) as the science and engineering of making intelligent machines, especially intelligent computer programs. Much earlier, the birth of AI was marked by Alan Turing's seminal paper 'Computing Machinery and Intelligence' (1950). Turing — often called the father of computer science — raised the question 'Can machines think?' and later proposed the Turing Test, in which a human interrogator tries to distinguish a computer from a human through text responses.

Popular languages used to code AI applications include R, Python and Java.

From a layman's view, AI is the intelligence demonstrated by machines that helps them mimic human actions. AI simulates natural intelligence in machines programmed to learn from experience, adapt to new inputs and perform human-like tasks.

From a researcher's view, AI is a set of algorithms that produces results without being explicitly instructed, making machines capable of thinking and acting rationally and humanely. Examples range from chess-playing computers to self-driving cars, relying heavily on deep learning and natural-language processing.

NITI Aayog (National Strategy for Artificial Intelligence) defines AI as the ability of machines to perform cognitive tasks like thinking, perceiving, learning, problem solving and decision making. Initially conceived as a technology that could mimic human intelligence, AI has since evolved well beyond its original conception.

Table 1.1 Definitions of Artificial Intelligence

Viewpoint	Meaning / Emphasis
Layman	Intelligence demonstrated by machines; mimics human actions; learns and adapts.
Researcher	A set of algorithms producing results without explicit instruction; rational, humane thinking and acting.
NITI Aayog	Ability of machines to perform cognitive tasks — thinking, perceiving, learning, problem solving, decision making.

Bloom's Taxonomy Questions — 1.1 What is Artificial Intelligence?

- 1 **L1** Define Artificial Intelligence according to John McCarthy.
- 2 **L2** Explain the meaning of AI from a layman's view and from a researcher's view.
- 3 **L3** Identify three real-world applications where machines perform tasks that normally require human intelligence.
- 4 **L4** Differentiate between human intelligence and machine intelligence based on perception, learning and decision-making.
- 5 **L5** Judge whether the Turing Test is a sufficient method to determine machine intelligence. Give reasons.
- 6 **L6** Design a simple AI-based classroom assistant and list the intelligent tasks it should perform.

GATE / Placement-Practice Questions — 1.1 What is AI?

Each item is tagged with a Bloom level (L1-L6); the answer/hint follows in *italics*.

- 1 **L1** The term 'Artificial Intelligence' was coined by: (A) Alan Turing (B) John McCarthy (C) Allen Newell (D) Marvin Minsky — (B) John McCarthy, at the 1956 Dartmouth Conference.
- 2 **L2** Which best describes the Turing Test? (A) a CPU benchmark (B) a test of whether a machine's responses are indistinguishable from a human's (C) a chess championship (D) a vision benchmark — (B).
- 3 **L3** For a loan-approval AI, which cognitive task is most directly used? (A) decision making (B) locomotion (C) musical ability (D) none — (A) decision making.
- 4 **L4** Statement I: computers outperform humans at sorting/memorising. Statement II: computers match humans at recognising emotion. (A) both true (B) I true, II false (C) I false, II true (D) both false — (B).
- 5 **L5** A chatbot passes the Turing Test 60% of the time. Does this prove human-like intelligence? Justify. — No — it tests behavioural imitation, not understanding (cf. Chinese Room argument).
- 6 **L6** Propose three measurable metrics for an AI classroom assistant beyond passing the Turing Test. — Answer accuracy, latency, student satisfaction, hallucination/error rate.

Previous Year VTU Questions — 1.1 What is AI?

Curated from VTU B.E./B.Tech papers (AI / Machine Learning / AI & Applications). Marks and scheme indicated where known.

- Define Artificial Intelligence. Explain the different views of defining AI with examples. (Jun-Jul 2024, 8 marks)
- What is AI? Briefly explain the foundations and applications of AI. (Jan 2023, 6 marks)
- Describe the Turing Test and discuss its significance in measuring machine intelligence. (Dec 2022, 8 marks)
- Write a short note on the NITI Aayog definition of AI and its national strategy. (Jun 2023, 6 marks)

1.1.1 How Does AI Work?

AI systems work effectively when fed large volumes of labelled training data. The data is analysed to discover correlations and patterns, which are then used to make predictions about future states. For example, a chatbot fed examples of text chats can learn to converse with humans. AI programming focuses on three cognitive skills:

- **Learning** — acquiring data and creating rules (algorithms) that turn data into actionable information.
- **Reasoning** — choosing the right algorithm to reach the desired outcome.
- **Self-correction** — continually refining algorithms to deliver the most accurate results.

Case study — Uber uses sophisticated ML algorithms to predict where riders will need taxis, dispatching drivers before demand spikes.

Bloom's Taxonomy Questions — 1.1.1 How Does AI Work?

- 1 **L1** List the three cognitive skills focused on in AI programming.
- 2 **L2** Explain how labelled training data helps an AI system discover patterns and make predictions.
- 3 **L3** Use the chatbot example to describe how an AI system learns to converse with users.
- 4 **L4** Break down the roles of learning, reasoning and self-correction in an AI system.
- 5 **L5** Assess the importance of choosing the right algorithm for an AI application.
- 6 **L6** Propose a workflow for an AI app that predicts student performance from marks and attendance.

GATE / Placement-Practice Questions — 1.1.1 How AI Works

Each item is tagged with a Bloom level (L1-L6); the answer/hint follows in *italics*.

- 1 **L1** The three cognitive skills of AI programming are learning, reasoning and _____. — *self-correction.*
- 2 **L2** Why is labelled training data important in supervised AI? — *Labels provide ground truth to learn input-output mappings and correct errors.*
- 3 **L3** A spam filter updates rules after misclassification. Which cognitive skill is this? — *Self-correction.*
- 4 **L4** Order the steps: (a) choose algorithm (b) collect labelled data (c) predict (d) find patterns. — *(b)→(d)→(a)→(c).*
- 5 **L5** If two algorithms have equal training accuracy, what do you evaluate before deployment? — *Generalisation, latency, interpretability, cost, fairness.*
- 6 **L6** Draft a 4-step pipeline for an attendance+marks predictor. — *Data collection → feature engineering → model selection/training → prediction + feedback retraining.*

Previous Year VTU Questions — 1.1.1 How Does AI Work?

Curated from VTU B.E./B.Tech papers (AI / Machine Learning / AI & Applications). Marks and scheme indicated where known.

- Explain how an AI system works with learning, reasoning and self-correction processes. (Jun 2024, 8 marks)
- Describe with a case study how a data-driven AI system creates business value. (Jan 2023, 6 marks)

1.1.2 Advantages and Disadvantages of AI

AI, used with machine learning and deep learning, processes massive data faster and predicts more accurately than humans. In 2017, Google CEO Sundar Pichai declared Google an 'AI-first' company.

Table 1.2 Advantages and Disadvantages of AI

#	Advantages	#	Disadvantages
1	Performs well on data-intensive tasks	1	High development and infrastructure cost
2	Processes huge data volumes quickly	2	Requires technical expertise
3	Generates consistent, accurate results	3	Shortage of trained professionals
4	Available 24×7	4	Bad/incomplete data can cause disastrous results
5	Optimises resource use	5	Limited ability to generalise across tasks
6	Automates complex processes		
7	Predicts maintenance, minimising downtime		
8	Faster, higher-quality new products		

Bloom's Taxonomy Questions — 1.1.2 Advantages and Disadvantages

- L1** List any four advantages of Artificial Intelligence.
- L2** Explain why AI applications can process huge volumes of data faster than humans.
- L3** Give an example of how AI can reduce downtime in an organisation.
- L4** Analyse how incomplete or inaccurate data can lead to wrong AI predictions.
- L5** Evaluate whether the benefits of AI outweigh its disadvantages in business.
- L6** Develop a short plan for implementing AI in a library, with benefits and risks.

GATE / Placement-Practice Questions — 1.1.2 Advantages / Disadvantages

Each item is tagged with a Bloom level (L1-L6); the answer/hint follows in *italics*.

- 1 **L2** Which is NOT an advantage? (A) 24×7 operation (B) consistent results (C) zero cost (D) automation — (C) zero cost.
- 2 **L3** Predictive maintenance reducing downtime best represents which advantage? — Proactive failure prediction.
- 3 **L4** A hiring model trained on male-dominated records rejects qualified women. Which disadvantage? — Bias from unrepresentative data.
- 4 **L5** A 98%-accurate but opaque diagnostic AI: list two benefits and two risks. — Benefits: speed/consistency; risks: explainability/accountability, bias, privacy.
- 5 **L6** Design a risk-mitigation plan for an AI library system. — Data validation, audit logs, librarian review, opt-out, periodic retraining.

Previous Year VTU Questions — 1.1.2 Advantages and Disadvantages

Curated from VTU B.E./B.Tech papers (AI / Machine Learning / AI & Applications). Marks and scheme indicated where known.

- List and explain the advantages and disadvantages of AI. (Jul 2024, 10 marks)
- Explain four advantages and four disadvantages with real-time examples. (Jan 2024, 8 marks)

1.2 History of Artificial Intelligence

Let us trace the modern field of AI from 1943 to the present; the principal milestones are summarised in Table 1.3.

Table 1.3 Milestones in the History of AI

Year	Milestone	Significance
1943	McCulloch & Pitts neural-network model	First mathematical model of a neural network.
1950	Turing Test; Minsky & Edmonds neural-network computer; Shannon chess paper; Asimov's Three Laws	Foundational questions of machine intelligence.
1952	Arthur Samuel self-learning checkers	Early self-learning program.
1956	McCarthy coins 'AI' at Dartmouth	Birth of AI; Newell & Simon demonstrate Logic Theorist.
1958	McCarthy develops LISP	Dominant AI programming language.
1959	Newell, Simon & Shaw build GPS; Samuel coins 'machine learning'	General Problem Solver; term ML born.

Year	Milestone	Significance
1963	Stanford AI Lab; Weizenbaum begins ELIZA	Early NLP / chatbot foundation.
1974–80	First AI Winter	DARPA cutbacks; funding dries up.
1980	DEC's R1/XCON expert system	First successful commercial expert system.
1982–93	Japan FGCS; Second AI Winter	Ambitious project terminated after ~\$1B spend.
1997	IBM Deep Blue beats Kasparov	Game-AI milestone.
2005	STANLEY wins DARPA Grand Challenge	Self-driving breakthrough.
2011	Apple Siri; IBM Watson wins Jeopardy!	Voice assistants and QA at scale.
2012	Google Brain recognises cats from YouTube	Deep-learning scale-up.
2014	Google self-driving car; Amazon Alexa	Consumer AI proliferation.
2016	AlphaGo defeats Lee Sedol; Sophia	Mastery of Go; humanoid robotics.
2018	Google BERT; Waymo One	NLP advances; commercial self-driving.
2020	Baidu LinearFold for COVID-19 vaccine	AI for rapid scientific response.

Bloom's Taxonomy Questions — 1.2 History of AI

- 1 **L1** Recall the year John McCarthy coined the term 'Artificial Intelligence'.
- 2 **L2** Explain the significance of Alan Turing's contribution.
- 3 **L3** Arrange ELIZA, Deep Blue, Siri, AlphaGo and BERT chronologically.
- 4 **L4** Analyse the causes and effects of the First and Second AI Winters.
- 5 **L5** Evaluate the importance of Deep Blue defeating Kasparov.
- 6 **L6** Create a timeline of at least eight AI milestones.

GATE / Placement-Practice Questions — 1.2 History

Each item is tagged with a Bloom level (L1-L6); the answer/hint follows in *italics*.

- 1 **L1** In which year was 'AI' coined? (A) 1950 (B) 1956 (C) 1969 (D) 1980 — (B) 1956.
- 2 **L1** Which AI language did McCarthy develop in 1958? (A) PROLOG (B) LISP (C) Python (D) Java — (B) LISP.
- 3 **L2** The First AI Winter was caused primarily by: — Funding cuts after over-promised, under-delivered research.
- 4 **L3** Chronology of ELIZA, Deep Blue, AlphaGo, BERT: — ELIZA(1966)→Deep Blue(1997)→AlphaGo(2016)→BERT(2018).
- 5 **L4** What pattern links both AI Winters? — Over-optimism and unmet expectations leading to funding withdrawal.
- 6 **L5** Why is AlphaGo considered a bigger breakthrough than Deep Blue? — Go has $\sim 10^{170}$ positions, requiring deep RL/neural methods rather than brute force.

Previous Year VTU Questions — 1.2 History

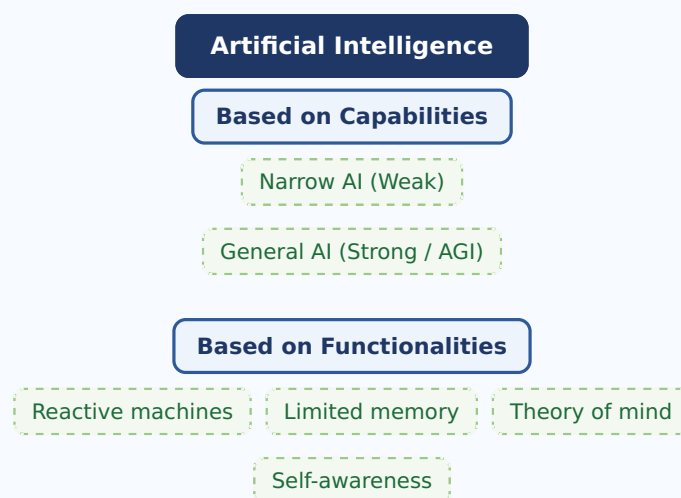
Curated from VTU B.E./B.Tech papers (AI / Machine Learning / AI & Applications). Marks and scheme indicated where known.

- Explain the brief history of AI with key milestones. (Jun 2024, 10 marks)
- What is an AI Winter? Discuss causes and consequences of both winters. (Jan 2023, 8 marks)
- Write a note on the evolution of AI from the Turing Test to AlphaGo. (Dec 2022, 8 marks)

1.3 Types of Artificial Intelligence

An AI system is classified by capability (Narrow vs. General/Strong AI) and by functionality (Reactive machines, Limited memory, Theory of mind, Self-aware), as shown in Figure 1.2.

Figure 1.2 Classification of Artificial Intelligence



1.3.1 Weak AI (Narrow AI)

Weak or narrow AI performs a specific task within a limited context — Siri, Alexa, Google Search, image recognition, self-driving cars and IBM Watson. It is the most successful realisation of AI to date.

1.3.2 Strong AI (AGI / ASI)

Strong AI — Artificial General Intelligence (AGI) or Superintelligence (ASI) — aims to resemble the human brain, using cognitive skills and fuzzy logic on tasks it was not trained for. It would require perception, speech recognition, decision-making and translation; today it exists mainly in science fiction.

Table 1.4 Weak AI vs. Strong AI

Basis	Weak AI (Narrow)	Strong AI (AGI/ASI)
Scope	Narrow range, limited scope	Wide range, broad scope
Capability	Good at specific tasks	Human-level (or greater) general intelligence
Techniques	Supervised / unsupervised learning	Clustering, association, cognitive/fuzzy reasoning
Examples	Siri, Alexa, Google Search, Watson	Advanced robotics (aspirational); HAL in sci-fi

1.3.3 Reactive Machines

Reactive machines have no memory; they perceive the world and react to the current state. Deep Blue and AlphaGo are examples. They respond identically to the same stimuli but cannot improve with practice or plan ahead.

1.3.4 Limited Memory

Limited-memory systems retain recent data briefly to inform decisions — essential in autonomous vehicles. Three ML models apply limited memory: reinforcement learning, Long Short-Term Memory (LSTM), and Evolutionary GANs (E-GAN).

1.3.5 Theory of Mind

Theory-of-mind AI would form representations of thoughts, emotions and memories to understand behaviour; it is currently theoretical.

1.3.6 Self-Awareness

Self-aware AI would possess human-level consciousness, understand its own existence and infer others' feelings. Such systems do not yet exist. The four functionality-based types are compared in Table 1.5.

Table 1.5 AI Systems Based on Functionality (Reactive, Limited Memory, Theory of Mind, Self-Aware)

Reactive AI	Limited Memory	Theory of Mind	Self-aware
Good for simple classification / pattern recognition; cannot handle imperfect information (current state of AI).	Handles complex classification; uses historical data to predict; powers self-driving cars but vulnerable to outliers/adversarial examples.	Understands human motives/reasoning; delivers personalised experience; can learn from fewer examples.	Human-level intelligence that can surpass ours; the next milestone in AI evolution.

Bloom's Taxonomy Questions — 1.3 Types of AI

- L1** Name the major types of AI based on capabilities and functionalities.
- L2** Explain the difference between Weak AI and Strong AI.
- L3** Classify Siri, Alexa, Deep Blue, AlphaGo and self-driving cars.
- L4** Compare reactive machines and limited-memory machines with examples.
- L5** Justify why self-awareness is more advanced than theory of mind.
- L6** Design a comparison table of all six categories.

GATE / Placement-Practice Questions — 1.3 Types

Each item is tagged with a Bloom level (L1-L6); the answer/hint follows in *italics*.

- L1** Siri and Alexa are examples of: (A) Strong AI (B) Reactive (C) Narrow/Weak AI (D) Self-aware — (C).
- L2** Which type uses recent observations to decide? (A) Reactive (B) Limited memory (C) Theory of mind (D) Self-aware — (B).
- L3** Deep Blue has no memory and responds to the current board; it is a ____ machine. — reactive.
- L4** Why are self-driving cars 'limited memory'? — They briefly store nearby cars' speeds/distances to navigate.
- L4** Match (1) Deep Blue (2) AGI (3) autonomous car — (a) limited memory (b) reactive (c) strong AI — 1-b, 2-c, 3-a.
- L5** Why is AlphaGo more advanced than Deep Blue? — Neural networks/RL evaluate future moves; Deep Blue is reactive brute force.
- L6** Place a medical-diagnosis AI on the weak/strong and functionality scales. — Today narrow + limited memory; roadmap toward theory-of-mind.

Previous Year VTU Questions — 1.3 Types

Curated from VTU B.E./B.Tech papers (AI / Machine Learning / AI & Applications). Marks and scheme indicated where known.

- Explain the types of AI based on capabilities with examples. (Jul 2024, 8 marks)
- Describe the four functionality-based types of AI. (Jan 2024, 10 marks)
- Differentiate weak AI and strong AI with examples. (Jun 2023, 6 marks)

1.4 Is AI the Same as Augmented Intelligence and Cognitive Computing?

AI is not the same as augmented intelligence. Augmented intelligence is weak AI that improves products and services — e.g., automatically highlighting vital business information. True AI (strong AI/AGI) would far surpass the human brain; quantum computing may help realise AGI. Cognitive computing describes products that mimic/augment human thought. When applied to machines, 'AI' means machines acquiring human-like intelligence by simulating sensing, learning, processing and reacting.

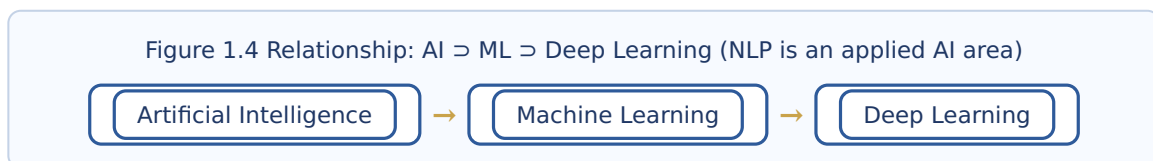
Table 1.6 AI vs. Augmented Intelligence vs. Cognitive Computing

Term	Core Idea	Examples
Artificial Intelligence (True/Strong AI)	Machines that obtain human-like intelligence for complex tasks.	AGI/ASI (aspirational)
Augmented Intelligence	Weak AI that assists/improves human decisions and products.	Highlighting key business info
Cognitive Computing	Systems that mimic/augment human thought.	Watson-style assistants
Machine Learning	Subset of AI that learns from data without explicit programming.	Spam filters, price prediction

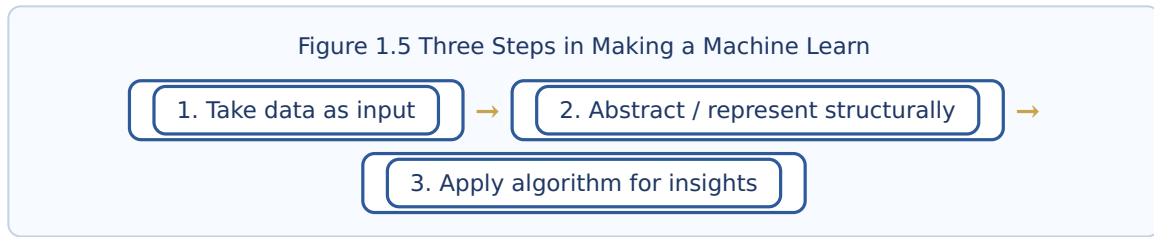
Machine learning through scenarios:

- Missing-number series 10, 20, 30, 40, ? — we observe the pattern and infer 50; ML imbibes this in machines.
- A cricketer uses past experience of a bowler; ML makes machines learn from past data.
- Given a phone configuration, a person estimates its price; ML predicts values from specifications.

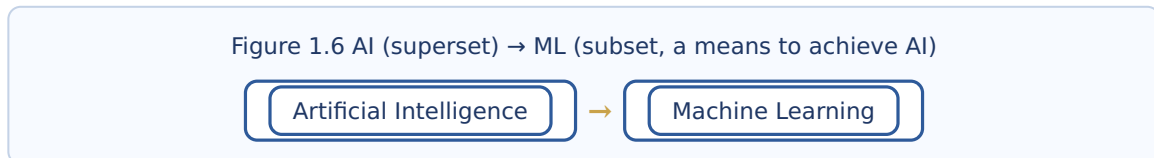
Machine Learning analyses data, identifies patterns and teaches a machine to decide without human intervention. It learns from experience rather than instructions, improving automatically. ML is an application (subset) of AI; Deep Learning is a subset of ML, and NLP is one of AI's applied areas (Figure 1.4).



Teaching a machine follows three steps (Figure 1.5): (1) take data as input (files, spreadsheets, databases); (2) abstract/represent the data structurally; (3) apply an appropriate algorithm to gain insights.



As Figure 1.6 summarises: AI builds intelligent machines that think/act like humans, while ML is the technique that lets systems learn without being explicitly programmed.



Traditional Programming vs. Machine Learning

Traditional programming: Input + Program = Output, with rules hand-coded in C/C++/Java/Python. Machine learning: Input + Output = Program — the algorithm itself formulates rules (a predictive model), e.g., a churn model from demographics/transactions plus historical churn labels.

Table 1.7 Traditional Programming vs. Machine Learning

Aspect	Traditional Software	Machine Learning
Objective	Meet functional/non-functional requirements	Optimise model performance; 0.1% gain matters
Quality depends on	Quality of code	Data quality and hyperparameter tuning
Built using	Hand-coded rules (C/C++/Java)	Different algorithms, each performing differently
Parameters	Few	Many
Change in data	New rules may be added	Model may need retraining
Example	Code $F = C \cdot 1.8 + 32$	Feed C/F pairs; ML infers conversion

ML supplements rather than replaces traditional programming: ML builds predictions while traditional code builds UI/visualisation.

Bloom's Taxonomy Questions — 1.4 AI, Augmented, Cognitive & ML

- 1 **L1** Define augmented intelligence and cognitive computing.
- 2 **L2** Explain why augmented intelligence is not true AI.
- 3 **L3** Classify auto-highlighting vital business info as AI, augmented or cognitive.
- 4 **L4** Differentiate traditional programming and ML using input/program/output.
- 5 **L5** Evaluate: 'ML supplements traditional programming but does not replace it.'
- 6 **L6** Create a real ML example with historical input/output producing a predictive model.

GATE / Placement-Practice Questions — 1.4 AI vs Augmented/Cognitive; ML vs Traditional

Each item is tagged with a Bloom level (L1-L6); the answer/hint follows in *italics*.

- 1 **L1** Which is a subset of AI? (A) ML (B) AR (C) Blockchain (D) Cloud — (A) ML.
- 2 **L2** In ML, which two are supplied to produce the program? — Input data and known output/label.
- 3 **L3** Auto-highlighting report risks is best called: (A) AGI (B) augmented intelligence (C) self-aware (D) theory of mind — (B).
- 4 **L4** Contrast: Traditional = Input + ? = Output; ML = Input + ? = ? — Traditional: Input + Program = Output; ML: Input + Output = Program.
- 5 **L4** Which is TRUE? (A) ML never retrains (B) ML quality depends heavily on data (C) traditional code auto-learns (D) ML needs fewer params — (B).
- 6 **L5** A startup uses ML for fraud detection but hand-codes the dashboard. Justify. — ML predicts; code handles deterministic UI — each plays to its strength.
- 7 **L6** Design a churn-prediction setup: business question, input, output. — Will a customer churn? Input: demographics/transactions/tickets; Output: churned/not; model predicts risk.

Previous Year VTU Questions — 1.4 AI, Augmented, Cognitive & ML

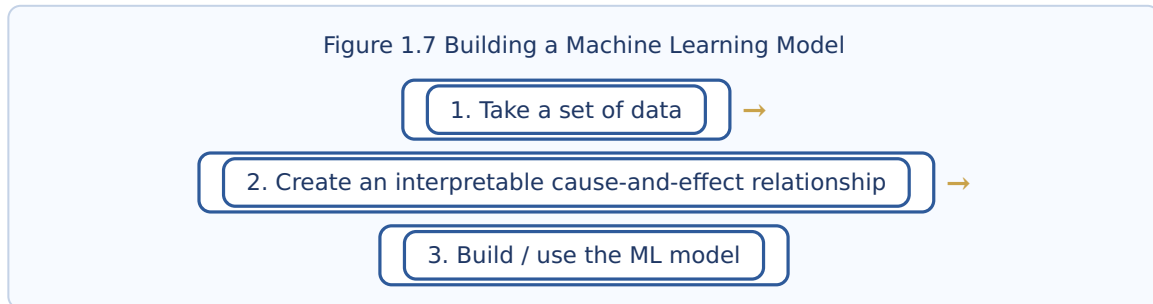
Curated from VTU B.E./B.Tech papers (AI / Machine Learning / AI & Applications). Marks and scheme indicated where known.

- Differentiate AI, augmented intelligence and cognitive computing. (Jun 2024, 8 marks)
- What is ML? Explain how machines learn with a diagram. (Jan 2024, 10 marks)
- Compare traditional programming and ML with an example. (Dec 2022, 8 marks)
- Explain the relationship between AI and ML with a diagram. (Jun 2023, 6 marks)

1.5 Machine Learning and Deep Learning

Frank Chen summarised: 'AI is a set of algorithms to mimic human intelligence. ML is one of them, and deep learning is one of those ML techniques.' ML feeds data to a computer so statistical techniques let machines learn without explicit programming. It is supervised (labelled data) or unsupervised (unlabelled).

Deep learning is an advanced ML technique that passes inputs through a biologically inspired, multi-layered neural network. Several hidden layers process data, letting machines go ‘deep’ — making connections and weighting inputs for the best results. AI algorithms solve tasks by studying large data and making statistical generalisations (Figure 1.7).



Features of AI:

1. Autonomous — decides without human intervention; learns from data and experience.
2. Predicts and adapts to data patterns.
3. Continuously learns from patterns.
4. Reactive — perceives and acts on perception.
5. Data-driven — cheap storage, fast CPUs/GPUs/TPUs and deep-learning algorithms enable data-centric AI.
6. Accurate predictions from past experience can outperform humans, given large correctly labelled datasets.
7. Futuristic — its scope keeps expanding.

Bloom's Taxonomy Questions — 1.5 ML and Deep Learning

- 1 **L1** Define ML and Deep Learning.
- 2 **L2** Explain how AI, ML, Deep Learning and NLP are related.
- 3 **L3** Use the Celsius-Fahrenheit example to contrast ML with traditional programming.
- 4 **L4** Analyse why deep learning uses multiple hidden layers.
- 5 **L5** Judge the importance of large labelled datasets.
- 6 **L6** Design an ML problem statement with question, input and output.

GATE / Placement-Practice Questions — 1.5 ML/DL

Each item is tagged with a Bloom level (L1-L6); the answer/hint follows in *italics*.

- 1 **L1** Deep learning is a subset of: (A) NLP (B) ML (C) Robotics (D) Expert systems — (B).
- 2 **L2** Supervised learning uses ____ data. — labelled.
- 3 **L3** A network with many hidden layers is a ____ network. — deep (deep-learning).
- 4 **L4** Why multiple hidden layers? — They learn hierarchical feature representations for better accuracy.
- 5 **L4** Order broad to narrow: Deep Learning, AI, ML. — $AI \supseteq ML \supseteq \text{Deep Learning}$.
- 6 **L5** 99% train accuracy but production failure — likely issue? — Overfitting/poor generalisation; check bias/variance and data representativeness.
- 7 **L6** Propose a DL book-recommender problem. — Inputs: history/genre/ratings; output: ranked recommendations; model learns user-item embeddings.

Previous Year VTU Questions — 1.5 ML and Deep Learning

Curated from VTU B.E./B.Tech papers (AI / Machine Learning / AI & Applications). Marks and scheme indicated where known.

- Differentiate ML and deep learning with examples. (Jul 2024, 8 marks)
- Explain the features of AI. (Jan 2023, 6 marks)
- Describe how an AI algorithm builds an ML model with a diagram. (Dec 2022, 8 marks)
- What is deep learning? Explain hidden layers. (Jun 2023, 8 marks)

PART B

Machine Intelligence

The nature of intelligence, rational agents and their environments, and the uninformed and informed search algorithms agents use to reach goals.

CHAPTER 3

The Artificially Intelligent Machine

3.1 Defining Intelligence

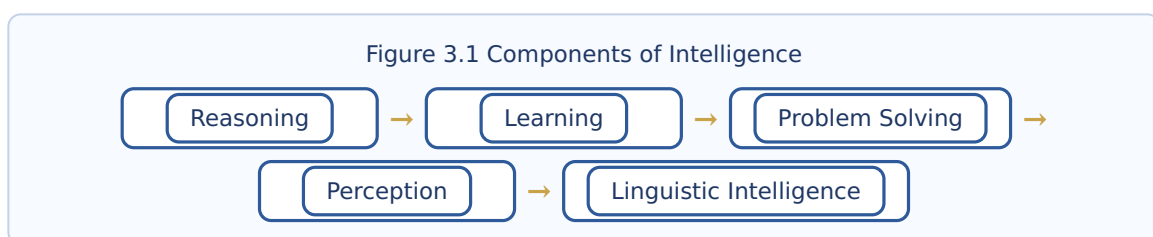
Howard Gardner categorised intelligence into several forms. A machine is artificially intelligent when it exhibits at least one (and at most all) of them.

Table 3.1 Gardner's Categories of Intelligence

Type	Description	Exhibited by
Linguistic	Use/recognise phonology, syntax and semantics.	Narrators, orators
Musical	Create and understand sounds, pitch and rhythm.	Musicians, composers
Logical-mathematical	Understand complex/abstract ideas.	Mathematicians, scientists
Spatial	Perceive and manipulate visual/spatial information.	Map readers, astronauts
Bodily-kinesthetic	Use body/body parts to solve problems or manipulate objects.	Players, dancers
Intrapersonal	Distinguish one's own feelings and motivations.	Self-aware individuals
Interpersonal	Recognise others' feelings, beliefs and intentions.	Communicators, interviewers

3.2 Components of Intelligence

Intelligence is composed of reasoning, learning, problem solving, perception and linguistic intelligence (Figure 3.1).



Reasoning

The set of processes used in making decisions and predictions — inductive or deductive.

Table 3.2 Inductive vs. Deductive Reasoning

Inductive Reasoning	Deductive Reasoning
Specific observations lead to broad general statements.	A general statement leads to a specific, logical conclusion.
Conclusion may be false even if observations are true.	If true for a class, it is true for all members.
'Manvi is a topper and studious; therefore all toppers are studious.'	'Students $\geq 80\%$ are eligible; Manan scored 90%; therefore Manan is eligible.'

Learning

Gaining knowledge/skill through study, practice, teaching or experience.

- Auditory — by listening (recorded lectures).
- Episodic — remembering sequences of events.
- Motor — precise muscle movement (picking objects).
- Observational — watching and imitating others.
- Perceptual — recognising previously seen stimuli.
- Relational — differentiating stimuli by relational properties.
- Spatial — through visual stimuli (maps, images).
- Stimulus-response — acting on a stimulus.

Problem solving

Finding a desired solution by selecting the best alternative path blocked by hurdles.

Perception

Acquiring, interpreting, selecting and organising sensory information; humans use sense organs, AI systems use sensors.

Linguistic intelligence

The ability to use, comprehend, speak and write verbal and written language.

3.3 Differences Between Human and Machine Intelligence

Table 3.3 Human vs. Machine Intelligence

#	Human Intelligence	Machine Intelligence
1	Perceives by patterns.	Perceives by analysing data against rules.
2	Stores/recalls by patterns.	Uses search algorithms to store and recall.

#	Human Intelligence	Machine Intelligence
3	Can deduce missing or distorted information.	Lacks high-accuracy deduction of missing/distorted info.

Bloom's Taxonomy Questions — 3.1-3.3 Intelligence and Its Components

- 1 **L1** List Gardner's categories of intelligence.
- 2 **L2** Explain reasoning, learning, problem solving, perception and linguistic intelligence.
- 3 **L3** Give examples of auditory, motor, observational and spatial learning.
- 4 **L4** Differentiate inductive and deductive reasoning with examples.
- 5 **L5** Evaluate whether machines can exhibit all forms of human intelligence.
- 6 **L6** Construct a concept map of intelligence components and their role in AI.

GATE / Placement-Practice Questions — 3.1-3.3 Intelligence

Each item is tagged with a Bloom level (L1-L6); the answer/hint follows in *italics*.

- 1 **L1** How many intelligences did Gardner propose (as listed)? (A) 3 (B) 5 (C) 7 (D) 9 — (C) 7.
- 2 **L2** 'All observed dogs bark; therefore all dogs bark' is ____ reasoning. — inductive.
- 3 **L3** 'All men are mortal; Socrates is a man; therefore Socrates is mortal' is: — deductive.
- 4 **L4** Match (1) auditory (2) motor (3) spatial — (a) reading a map (b) a lecture (c) catching a ball — 1-b, 2-c, 3-a.
- 5 **L5** Humans infer missing info better than machines today. Which component is this? — Reasoning + perceptual pattern completion.
- 6 **L6** Design a campus-navigation agent combining linguistic, logical and spatial intelligence. — NLP query understanding, route reasoning, map/visual perception.

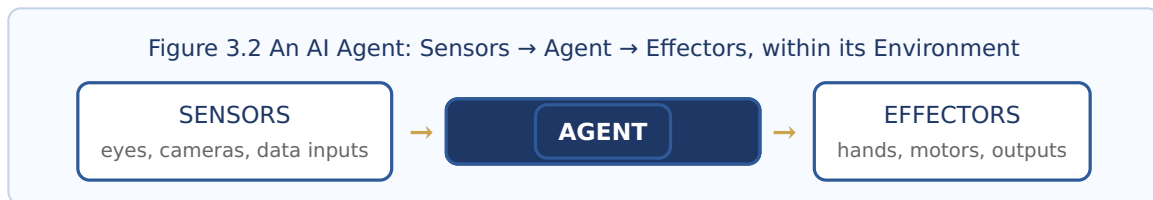
Previous Year VTU Questions — 3.1-3.3 Intelligence

Curated from VTU B.E./B.Tech papers (AI / Machine Learning / AI & Applications). Marks and scheme indicated where known.

- Explain Gardner's categories of intelligence. (Jul 2024, 8 marks)
- What is intelligence? Explain its components with a diagram. (Jan 2024, 10 marks)
- Differentiate inductive and deductive reasoning with examples. (Jun 2023, 6 marks)
- Explain the types of learning in intelligent systems. (Dec 2022, 8 marks)
- Compare human and machine intelligence. (Jun 2024, 6 marks)

3.4 Agent and Environment

AI is the study of rational agents — any decision maker (person, firm, machine, software) that acts for the best outcome given past and current percepts. An AI system comprises an agent and its environment; the agent perceives via sensors and acts via effectors (Figure 3.2).



As Figure 3.2 shows, sensors feed percepts to the agent and effectors carry actions back into the environment.

Table 3.4 Agent Types and Their Sensors/Effectors

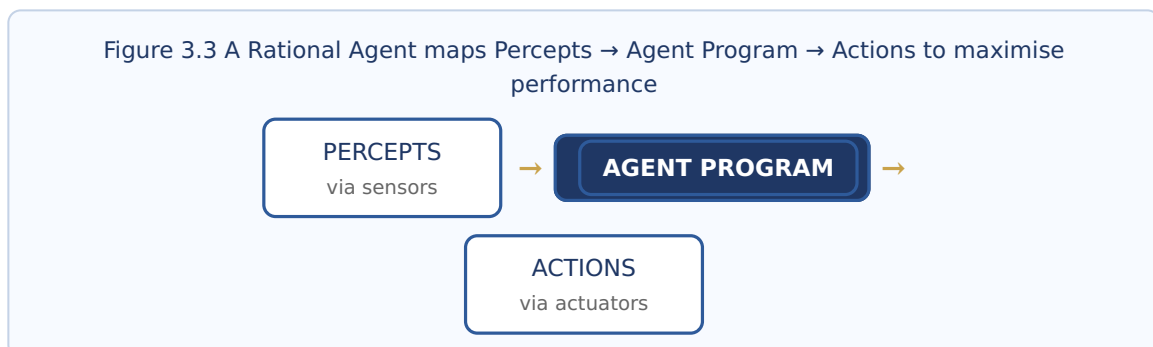
Agent Type	Sensors	Effectors
Human	Eyes, ears, nose, tongue, skin	Hands, legs, mouth
Robotic	Cameras, infrared range finders	Motors, actuators
Software	Bit strings (data inputs)	Bit strings (program actions)

3.4.1 Key Terminology

- **Performance measure** — criterion of agent success.
- **Behaviour** — action after receiving a percept.
- **Percept** — perceptual input at a specific instant.
- **Percept sequence** — all percepts received to date.
- **Agent function** — a mapping from percept sequence to action.

3.4.2 Rationality

Rationality empowers an agent to act to maximise its performance measure given its percept sequence and knowledge. It depends on the performance measure, percept sequence so far, prior environment knowledge, and available actions. A problem is described by PEAS — Performance, Environment, Actuators, Sensors. Structure: Agent = Architecture + Agent Program. Figure 3.3 shows how percepts flow in and actions flow out.



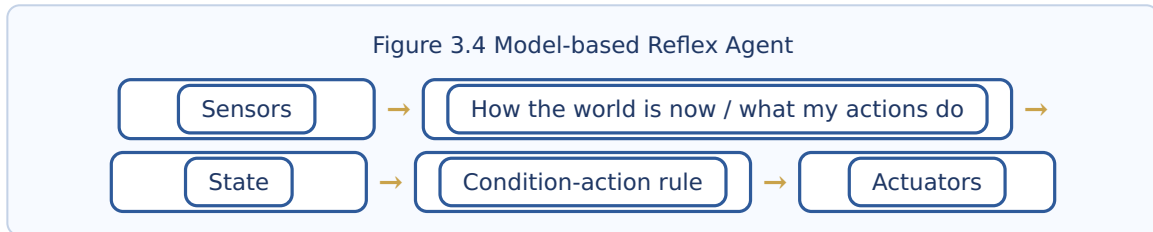
3.4.3 & 3.4.4 Structure and Types of Agents

Simple reflex agents

Act only on the current percept using condition-action rules; rational only in fully observable environments, and may loop in partially observable ones.

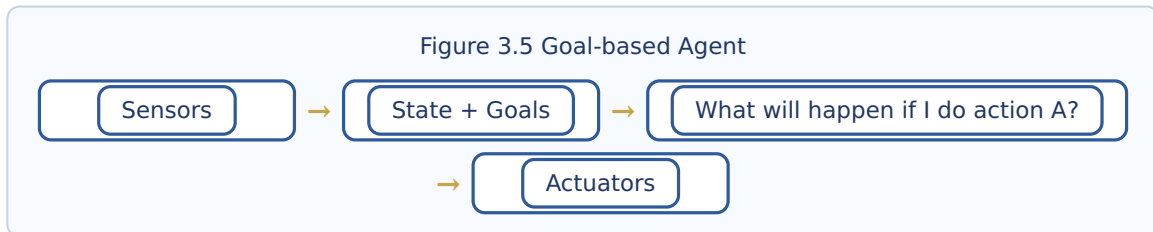
Model-based reflex agents

Maintain an internal state/model of how the world evolves and how actions affect it (Figure 3.4).



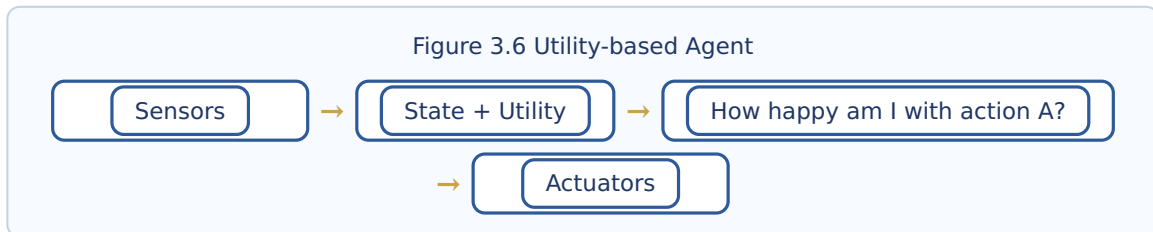
Goal-based agents

Choose actions to achieve explicit goals; flexible because decision knowledge is represented (Figure 3.5).



Utility-based agents

When goals conflict, choose actions that maximise expected utility ('happiness') (Figure 3.6).



Learning agents

Learn from experience through four components: learning element, critic (feedback against a standard), performance element and problem generator (Figures 3.7 and 3.8).

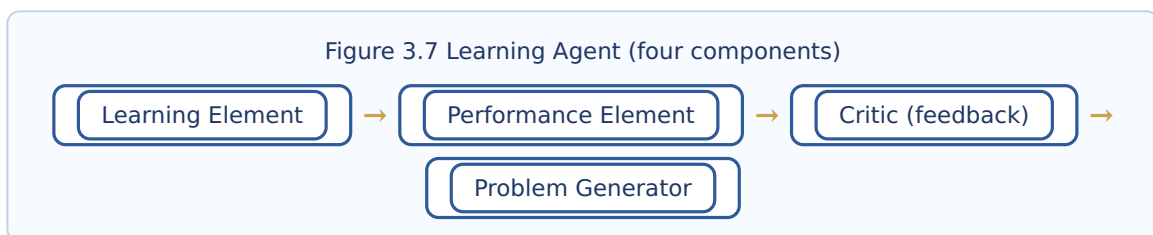


Figure 3.8 The Critic feeds performance feedback back to the Learning Element



Bloom's Taxonomy Questions — 3.4 Agents, Rationality and Types

- 1 **L1** Define agent, environment, sensor, effector, percept and agent function.
- 2 **L2** Explain Agent = Architecture + Agent Program.
- 3 **L3** Identify sensors and effectors for human, robotic and software agents.
- 4 **L4** Compare simple reflex, model-based, goal-based, utility-based and learning agents.
- 5 **L5** Justify why a rational agent maximises performance using percept sequence and prior knowledge.
- 6 **L6** Design a PEAS description for a smart vacuum cleaner or autonomous taxi.

GATE / Placement-Practice Questions — 3.4 Agents & Rationality

Each item is tagged with a Bloom level (L1-L6); the answer/hint follows in *italics*.

- 1 **L1** An agent's ____ maps a percept sequence to an action. — agent function.
- 2 **L1** PEAS stands for: — Performance, Environment, Actuators, Sensors.
- 3 **L2** A simple reflex agent is rational only when the environment is: (A) dynamic (B) fully observable (C) stochastic (D) multi-agent — (B).
- 4 **L3** For an autonomous taxi, list actuators and sensors. — Actuators: steering/accelerator/brake/signals; Sensors: cameras/lidar/GPS/speedometer.
- 5 **L4** Which agent maintains internal state for partial observability? — Model-based reflex agent.
- 6 **L4** Order by capability: goal-based, simple reflex, learning, utility-based. — simple reflex → goal-based → utility-based → learning.
- 7 **L5** Why can a utility agent outperform a goal agent with conflicting goals? — It quantifies preference and picks the best trade-off, not just any goal-achieving action.
- 8 **L6** Write PEAS for a smart vacuum and choose an agent type. — P: cleanliness/efficiency; E: rooms/furniture/pets; A: wheels/brush/suction; S: bump/dirt sensors; model-based + learning.

Previous Year VTU Questions — 3.4 Agents

Curated from VTU B.E./B.Tech papers (AI / Machine Learning / AI & Applications). Marks and scheme indicated where known.

- Define rational agent. Explain agent structure and the factors determining rationality. (Jul 2024, 10 marks)
- What is PEAS? Give the PEAS description for an autonomous taxi. (Jan 2024, 6 marks)
- Explain the different agent-program types with diagrams. (Dec 2022, 12 marks)
- Differentiate simple reflex, goal-based, utility-based and learning agents. (Jun 2023, 8 marks)
- Define agent, percept, percept sequence and agent function with a diagram. (Jun 2024, 8 marks)

3.4.5 The Nature of Environments

Some agents work in limited domains (keyboard input, databases); software robots (softbots) work in large, complex, real-time environments. The most famous artificial environment is the Turing Test: a tester questions a hidden human and machine through text; if the machine cannot be identified, it is deemed intelligent.

3.4.6 Types of Environments

Table 3.5 Dimensions of AI Environments (Figures 3.9–3.12 summarise these contrasts)

Dimension	Meaning	Example
Discrete / Continuous	Finite distinct states/actions vs. no such limit.	Chess (discrete); self-driving car (continuous).
Known / Unknown	Agent knows action results vs. must learn them.	Known-rules game; new robotics task.
Observable / Partial	Complete state from percepts vs. not.	Chess (fully); Kriegspiel (partial).
Static / Dynamic	Unchanged while acting vs. changing.	Crossword (static); self-driving car (dynamic).
Single / Multi-agent	One vs. several agents (competitive/cooperative).	Vacuum (single); chess (competitive).
Accessible / Inaccessible	Sensors access complete state vs. not.	Temperature room (accessible); world event (inaccessible).
Deterministic / Non-deterministic	Next state fixed by state+action vs. not; stochastic if probabilistic.	Chess; ludo (non-deterministic).

Dimension	Meaning	Example
Episodic / Sequential	Independent episodes vs. decisions affecting future.	Anomaly detection (episodic); chess (sequential).

Figure 3.9 (Observable/Partially observable): if sensors capture the complete relevant state at every instant, no internal state history is needed; noise, inaccurate sensors or missing data make it partially observable (e.g., Kriegspiel chess).

Figure 3.10 (Multi-agent): agent 1 and agent n act in a shared environment — competitive when performance is zero-sum (chess), cooperative when both must avoid collisions and maximise joint performance (two taxis).

Figure 3.11 (Deterministic/Non-deterministic): in a deterministic environment, next state = $f(\text{current state, action})$; non-deterministic environments (ludo, dice) add uncertainty, quantified as probabilities in stochastic environments.

Figure 3.12 (Episodic/Non-episodic): in an episodic environment each perceive-act episode is independent; in a sequential (non-episodic) environment current decisions affect all future decisions, so the agent needs memory.

Bloom's Taxonomy Questions — 3.4.5 & 3.4.6 Environments

- 1 **L1** List the different types of AI environments.
- 2 **L2** Explain fully vs. partially observable environments.
- 3 **L3** Classify chess, ludo, self-driving car and vacuum-cleaner environments.
- 4 **L4** Differentiate deterministic and non-deterministic environments with examples.
- 5 **L5** Evaluate why a self-driving car is dynamic and continuous.
- 6 **L6** Prepare a classification table for three AI applications.

GATE / Placement-Practice Questions — 3.4.5-3.4.6 Environments

Each item is tagged with a Bloom level (L1-L6); the answer/hint follows in *italics*.

- 1 **L1** Chess with all pieces visible best fits a(n) ____, ____ environment. — *fully observable, deterministic.*
- 2 **L2** An environment that changes while the agent deliberates is: (A) static (B) episodic (C) dynamic (D) discrete — *(C).*
- 3 **L3** Classify a self-driving car across the main dimensions. — *Partially observable, stochastic, sequential, dynamic, multi-agent, continuous.*
- 4 **L4** Ludo (dice) and poker (hidden cards) highlight which dimensions? — *Non-deterministic/stochastic and partially observable; multi-agent.*
- 5 **L5** Why is a static, fully observable, deterministic environment easier? — *No uncertainty or world change during action; planning is simpler.*
- 6 **L6** Build an environment-classification table for a medical-image diagnostic agent. — *Largely deterministic output, episodic per patient, static per image, single agent; discuss uncertainty.*

Previous Year VTU Questions — 3.4.5-3.4.6 Environments

Curated from VTU B.E./B.Tech papers (AI / Machine Learning / AI & Applications). Marks and scheme indicated where known.

- Explain the types of environments with examples. (Jul 2024, 10 marks)
- Differentiate observable/partial, deterministic/stochastic, static/dynamic and episodic/sequential environments. (Jan 2024, 10 marks)
- What is the Turing Test? How does it evaluate machine intelligence? (Dec 2022, 8 marks)
- Classify chess, self-driving car, vacuum cleaner and ludo with justification. (Jun 2023, 8 marks)

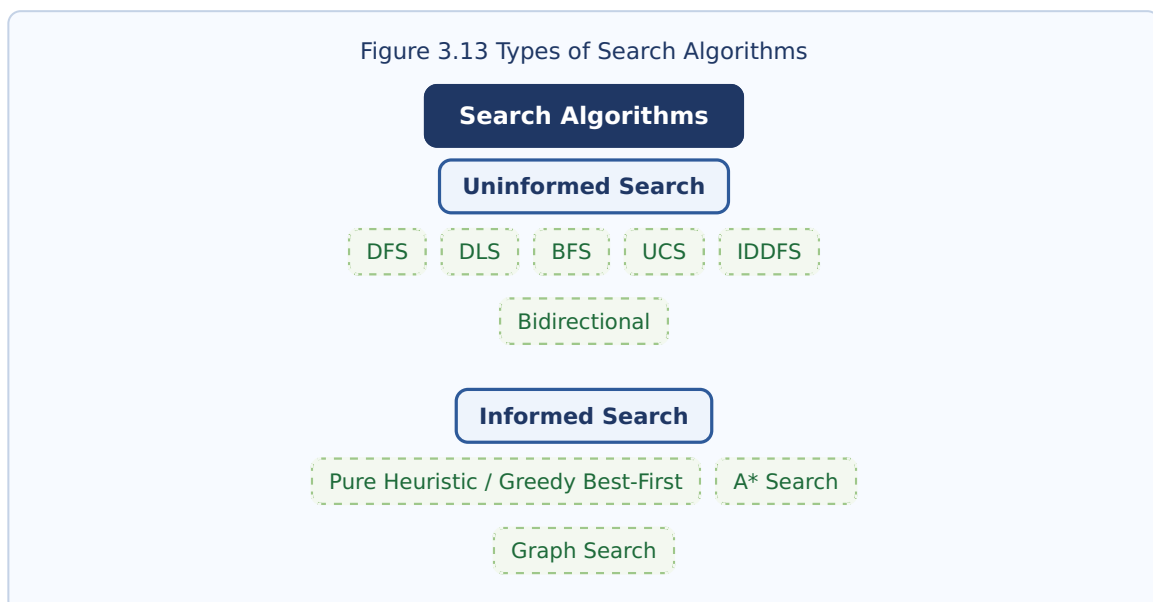
3.5 Search

Rational agents use search algorithms for tasks such as puzzles, Sudoku and crosswords. A search problem consists of:

- **State space** — all states attainable by the agent.
- **Start state** — where searching begins.
- **Goal test** — checks whether a state is the goal.
- **Solution (plan)** — action sequence from start to goal.

3.5.1 Types of Search Algorithms

The six popular algorithms are grouped into uninformed (blind) and informed (heuristic) search (Figure 3.13).



3.5.2 Properties of Search Algorithms

- **Completeness** — returns a solution if one exists.
- **Optimality** — finds the lowest-path-cost solution.
- **Time complexity** — time taken.

- **Space complexity** — maximum storage required.

Notation: b = branching factor; d = depth of shallowest solution; m = maximum depth; l = depth limit.

Bloom's Taxonomy Questions — 3.5 Search and Properties

- 1 **L1** Define state space, start state, goal test and plan.
- 2 **L2** Explain completeness, optimality, time and space complexity.
- 3 **L3** Identify start, goal and solution path for a simple route problem.
- 4 **L4** Analyse why search algorithms matter for rational agents.
- 5 **L5** Judge optimality vs. space complexity on a memory-limited system.
- 6 **L6** Create a small state-space diagram from classroom to library.

GATE / Placement-Practice Questions — 3.5 Search Fundamentals

Each item is tagged with a Bloom level (L1-L6); the answer/hint follows in *italics*.

- 1 **L1** A search algorithm is ____ if guaranteed to find a solution when one exists. — *complete.*
- 2 **L2** Branching factor b is: (A) goal depth (B) max successors of a node (C) total nodes (D) edge cost — *(B).*
- 3 **L3** For route finding, specify the goal test. — *'Current location == destination'.*
- 4 **L4** Why does an agent need search despite knowing the goal? — *To find an optimal action sequence/plan among alternatives.*
- 5 **L5** On an embedded device, would you choose a complete but space-hungry algorithm? — *Maybe not; trade completeness/optimality for bounded memory (DLS/IDDFS).*
- 6 **L6** Model classroom-to-library as states, actions, goal test, cost. — *States=locations; actions=corridor moves; goal=library; cost=distance/time.*

Previous Year VTU Questions — 3.5 Search

Curated from VTU B.E./B.Tech papers (AI / Machine Learning / AI & Applications). Marks and scheme indicated where known.

- Define a search problem. Explain state space, start state, goal test and solution. (Jul 2024, 8 marks)
- Explain completeness, optimality, time and space complexity. (Jan 2024, 6 marks)
- Differentiate uninformed and informed search. (Jun 2023, 6 marks)

3.6 Uninformed Search Algorithms

Uninformed (blind) search has no information beyond the problem definition; it only knows how to traverse nodes. Terms: problem graph ($S \rightarrow G$), strategy, fringe (reachable nodes), search tree, solution plan and step cost.

3.6.1 Depth-First Search (DFS)

DFS goes to the leaf of a branch, then backtracks; it uses a LIFO stack (Fig. 3.14). Time $O(b^m)$, space $O(b \cdot m)$; complete only on finite trees, not optimal.

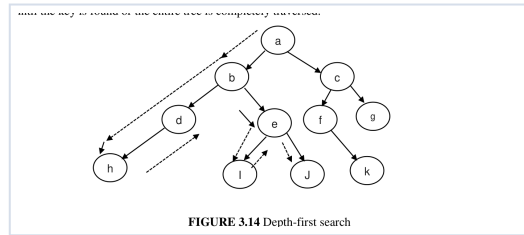


Figure 3.14 Depth-First Search traversal

- Advantages: low memory; faster to deep goals than BFS.
- Disadvantages: can loop infinitely; no duplicate check; cut-off depth may be needed.

3.6.2 Depth-Limited Search (DLS)

DLS is DFS with a depth limit l ; nodes at the limit are treated as leaves (Fig. 3.15). Time $O(b^l)$, space $O(b \cdot l)$; complete only if the solution is within l , and not optimal.

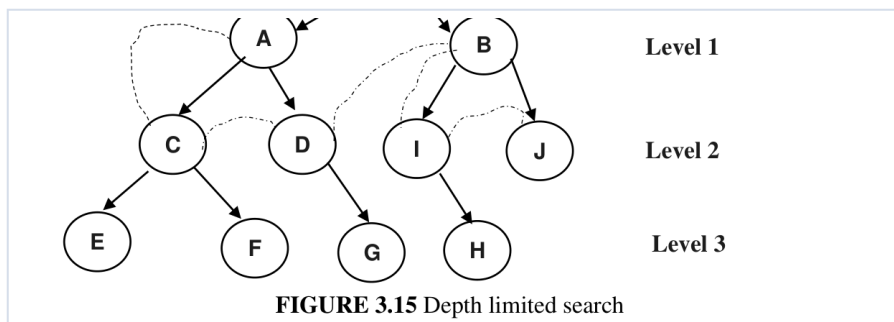


Figure 3.15 Depth-Limited Search

3.6.3 Breadth-First Search (BFS)

BFS expands level by level using a FIFO queue (Fig. 3.16, 3.17). Complete; optimal when step cost is 1; time/space $O(b^d)$. It finds shortest paths and is used in crawlers, GPS, broadcasting and P2P networks.

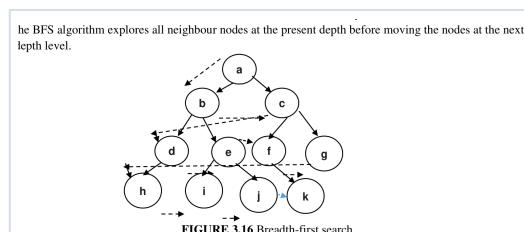


Figure 3.16 Breadth-First Search traversal

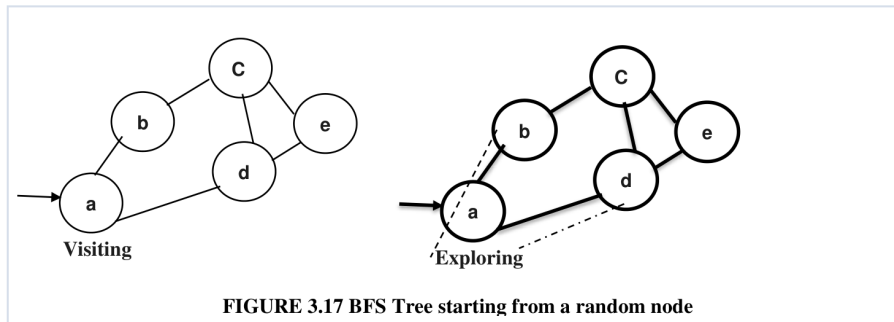


Figure 3.17 BFS on a graph — Visiting vs Exploring

```

BFS(G, s):
  Q = queue()
  Q.enqueue(s); mark s visited
  while Q is not empty:
    v = Q.dequeue()
    for each neighbour w of v:
      if w not visited:
        Q.enqueue(w); mark w visited

```

3.6.4 Uniform-Cost Search (UCS)

UCS expands the lowest cumulative path-cost node via a priority queue (Fig. 3.18); identical to BFS when all edge costs are equal. Complete (no zero-cost loops) and optimal (no negative edges). Example: $S \rightarrow A \rightarrow B \rightarrow G$ with cost 5.

Uniform Cost Search (UCS)

UCS algorithm is used to find an optimal solution to the goal state when the step costs are not equal. In such a scenario, the algorithm computes the cumulative cost to expand each node from the root to the goal node. It neither traverses depth nor breadth, but searches for the next node with the lowest cumulative cost. Sorting is done in increasing cost of the path to a node so that the algorithm can explore paths in increasing order of cost. UCS always expands the node with the least cost. It is identical to BFS if all edges have the same cost. Remember that, **Cost of a node** is defined as:

$\text{Cost}(\text{node}) = \text{cumulative cost of all nodes from root}$

$\text{Cost}(\text{root}) = 0$

Fig. 3.18, if S is the start node and G is the goal state, then node A is explored as it has the lowest cost (the other node G has a higher cost). Now, node A has two child nodes-B and C. C having the lowest step cost is traversed and from there, node D, followed by G is reached. Though UCS performs well for this algorithm, it may not work with all cases.

As we see that uniform-cost search (UCS) expands nodes based on their path costs from the root.

It can be used to traverse any graph/tree.

Where the optimal cost is the traversal criteria. The uniform-cost search algorithm is implemented by a priority queue, giving maximum priority to the lowest cumulative cost. If you observe carefully, you will find that UCS is equivalent to BFS algorithm if the path cost of all edges is the same.

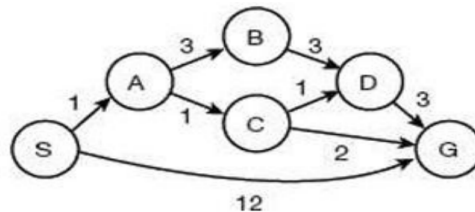


Figure 3.18 Uniform-Cost Search

3.6.5 Iterative Deepening DFS (IDDFS)

IDDFS runs DFS with increasing limits (1, 2, 3, ...) until the goal is found. Complete (if finite); optimal if cost is non-decreasing with depth; time $O(b^d)$, space $O(b \cdot d)$. Ideal for large spaces with unknown goal depth.

3.6.6 Bidirectional Search

Searches forward from start and backward from goal simultaneously until the frontiers meet; each side covers about half the path. Time/space $O(b^{(d/2)})$; complete and optimal with BFS and uniform costs. Requires a known goal and similar branching in both directions.

Table 3.6 Comparison of Uninformed Search Algorithms (* no zero-cost loops / negative edges)

Algorithm	Data Structure	Complete?	Optimal?	Time	Space
DFS	Stack	Finite tree only	No	$O(b^m)$	$O(b \cdot m)$

Algorithm	Data Structure	Complete?	Optimal?	Time	Space
DLS	Stack + limit	Within l only	No	$O(b^l)$	$O(b \cdot l)$
BFS	Queue	Yes	If cost=1	$O(b^d)$	$O(b^d)$
UCS	Priority queue	Yes*	Yes*	$O(b^{(1+\text{floor}(c^*/\epsilon)))})$	$O(b^{(1+\text{floor}(c^*/\epsilon)))})$
IDDFS	Stack	Yes (finite b)	If non-decreasing cost	$O(b^d)$	$O(b \cdot d)$
Bidirectional	Two queues	With BFS both ways	With uniform cost	$O(b^{(d/2)})$	$O(b^{(d/2)})$

Bloom's Taxonomy Questions — 3.6 Uninformed Search

- L1** List the uninformed search algorithms.
- L2** Explain DFS, DLS, BFS, UCS, IDDFS and bidirectional search.
- L3** Apply BFS and DFS to a tree and write the traversal order.
- L4** Compare BFS, DFS and UCS on completeness, optimality and complexity.
- L5** Evaluate why UCS is preferred for unequal path costs.
- L6** Design a weighted graph and demonstrate UCS least-cost search.

GATE / Placement-Practice Questions — 3.6 Uninformed Search (High GATE Weightage)

Each item is tagged with a Bloom level (L1-L6); the answer/hint follows in *italics*.

- L1** DFS uses a ____; BFS uses a _____. — stack; queue.
- L2** Which finds the shallowest goal in an unweighted graph? — BFS.
- L2** Which is optimal for unequal edge costs? — UCS (Dijkstra-style).
- L3** For $b=2$, $d=3$, worst-case BFS nodes? — $b+b^2+b^3 = 2+4+8 = 14$ (plus root).
- L3** Edges S-A(1), S-B(4), A-B(1), A-G(5), B-G(1): UCS path and cost? — S→A(1)→B(2)→G(3); cost 3.
- L4** Why is BFS memory a bigger practical concern than DFS? — BFS stores the whole frontier $O(b^d)$; DFS stores a path $O(bm)$.
- L4** IDDFS repeats work — why is it still efficient? — Repeated upper levels are geometrically small; space stays $O(bd)$.
- L5** If all edge costs are equal, UCS behaves like which algorithm? — BFS — priority by cost reduces to FIFO.
- L5** When is bidirectional search inappropriate? — Unknown goal, undefined predecessors, or very different branching factors.
- L6** For $b=10$, $d=6$, compare BFS vs bidirectional nodes. — BFS $\sim 10^6$; bidirectional $\sim 2 \cdot 10^3$ — about 500x fewer.

Previous Year VTU Questions — 3.6 Uninformed Search

Curated from VTU B.E./B.Tech papers (AI / Machine Learning / AI & Applications). Marks and scheme indicated where known.

- Explain DFS with a diagram; give time and space complexity. (Jul 2024, 10 marks)
- Explain BFS with an example and its merits/demerits. (Jan 2024, 10 marks)
- What is UCS? How does it differ from BFS? (Dec 2022, 8 marks)
- Explain DLS and IDDFS with examples. (Jun 2023, 10 marks)
- Explain bidirectional search and its complexity. (Jun 2024, 8 marks)
- Compare all six uninformed searches on completeness, optimality, time and space. (Jan 2023, 12 marks)

| 3.7 Informed Search Algorithms

Informed (heuristic) search uses extra knowledge — estimated distance, cost — to explore less and reach the goal faster. The heuristic $h(n)$ estimates the optimal cost from n to a goal and is positive. For an admissible heuristic, $h(n) \leq h^*(n)$ (it never overestimates).

3.7.1 Pure Heuristic Search

Expands nodes by $h(n)$, maintaining OPEN (unexpanded) and CLOSED (expanded) lists; each iteration expands the lowest- h node and inserts its children into OPEN until a goal is found.

3.7.2 Best-First Search (Greedy Search)

Greedy best-first search always expands the node with the smallest $h(n)$, using a priority queue. It can loop, is not optimal, and has worst-case $O(b^m)$. Worked example: $S \rightarrow B \rightarrow F \rightarrow G$.

1. Insert the start into OPEN.
2. If OPEN is empty, return failure.
3. Move the lowest- h node from OPEN to CLOSED.
4. Expand it and generate successors.
5. If a successor is the goal, return success.
6. Add successors to OPEN only if absent from OPEN/CLOSED.
7. Repeat.

Bloom's Taxonomy Questions — 3.7 Informed Search

- 1 **L1** Define informed search and heuristic function.
- 2 **L2** Explain how heuristic information improves search efficiency.
- 3 **L3** Choose the next node from an OPEN list using greedy search.
- 4 **L4** Analyse blind vs. heuristic search.
- 5 **L5** Evaluate the limitation of greedy search for optimality.
- 6 **L6** Create a route-search example with heuristic values.

GATE / Placement-Practice Questions — 3.7 Informed (Heuristic) Search

Each item is tagged with a Bloom level (L1-L6); the answer/hint follows in *italics*.

- 1 **L1** Greedy best-first selects the node with the lowest _____. — $h(n)$.
- 2 **L1** Which lists does pure heuristic search maintain? — OPEN and CLOSED.
- 3 **L2** An admissible heuristic satisfies $h(n)$ ____ $h^*(n)$. — $\leq$ (never overestimates).
- 4 **L2** Why is greedy not optimal? — It ignores cost-so-far $g(n)$ and chases low h , possibly taking costly paths/loops.
- 5 **L3** OPEN has $h(A)=5$, $h(B)=2$, $h(C)=7$. Greedy expands: — B ($h=2$).
- 6 **L4** Blind vs heuristic — trade-off? — Heuristic is faster but needs accurate domain knowledge; blind needs none but explores more.
- 7 **L5** What does A^* minimise that greedy ignores (preventing runaways)? — $f(n)=g(n)+h(n)$; $g(n)$ accounts for cost already incurred.
- 8 **L6** Define an admissible map heuristic and justify it. — Straight-line distance — roads are never shorter than straight-line, so $h \leq h^*$.

Previous Year VTU Questions — 3.7 Informed Search

Curated from VTU B.E./B.Tech papers (AI / Machine Learning / AI & Applications). Marks and scheme indicated where known.

- What is heuristic search? Explain the heuristic function and its properties. (Jul 2024, 8 marks)
- Explain greedy best-first search with algorithm and example. (Jan 2024, 10 marks)
- Differentiate uninformed and informed search. (Dec 2022, 6 marks)
- Explain pure heuristic search using OPEN/CLOSED with an example. (Jun 2023, 8 marks)

PART C

Knowledge Representation

How AI agents store, reason with and apply knowledge — what must be represented, the KR cycle, knowledge-based agents, and the types of knowledge used in intelligent systems.

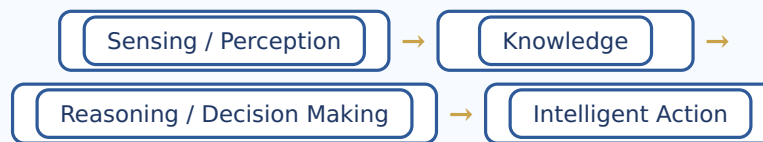
CHAPTER 4

Knowledge Representation

4.1 Introduction — Knowledge and Intelligence

To build AI systems with conscience, we must inculcate knowledge. Real-world knowledge is crucial for intelligent behaviour: an agent senses its environment and uses knowledge to act intelligently. Without knowledge, a decision-maker cannot sense accurately or choose appropriate actions (Figure 4.1).

Figure 4.1 Knowledge ↔ Intelligence: sensing and decision-making depend on knowledge



4.2 Knowledge Representation

Knowledge Representation and Reasoning (KR, KRR) is the part of AI concerned with how agents think and how thinking produces intelligent behaviour. A knowledge-based system represents real-world information so a computer can use it to solve complex problems. KR models intelligent behaviour and goes beyond database storage.

4.2.1 What Knowledge Needs to be Represented?

- **Objects** — facts about relevant objects (vehicles, roads).
- **Events** — actions that occur (applying brakes).
- **Performance** — how actions are performed.
- **Meta-knowledge** — knowledge about what we know.
- **Facts** — truths about the real world.
- **Knowledge base** — technical sentences; core of a KBA.

4.2.2 What is Knowledge?

Knowledge is gained through experience of facts, data and situations and helps us reason logically. The five types are listed in Table 4.1 and shown in Figure 4.2.

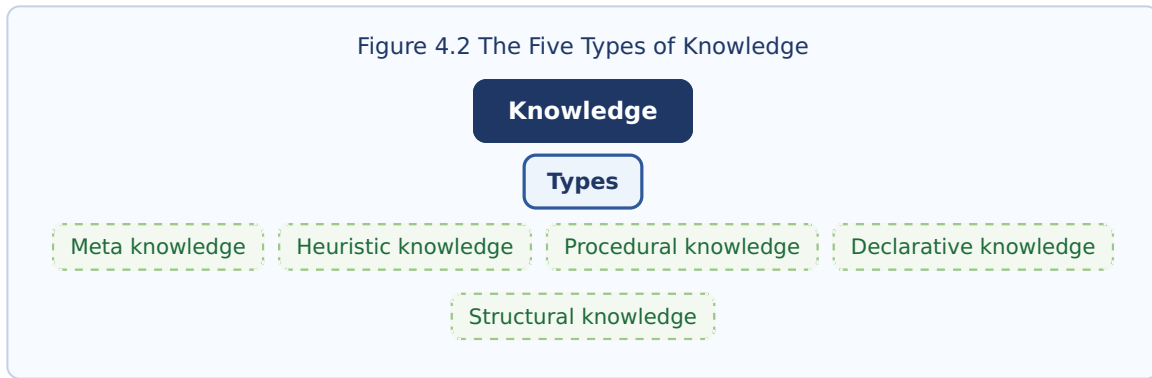


Table 4.1 Five Types of Knowledge

Type	Meaning
Meta knowledge	Knowledge about knowledge.
Heuristic knowledge	Rule-of-thumb experience; good but not guaranteed.
Procedural knowledge	How to do something — rules, strategies, procedures.
Declarative knowledge	Information about objects/concepts/facts, in declarative sentences.
Structural knowledge	Relationships such as kind-of/part-of/grouping.

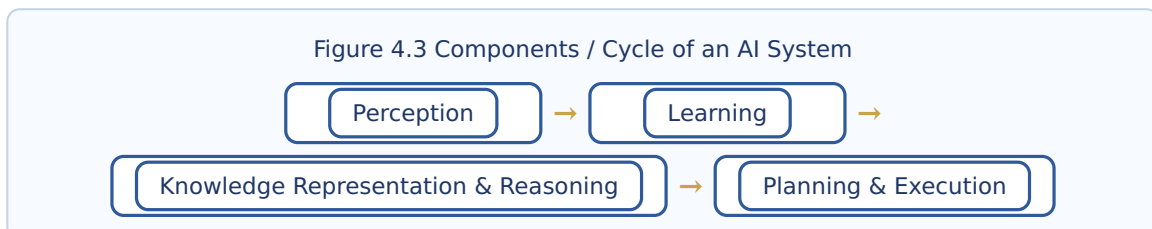
4.2.3 What is Logic?

Logic helps draw conclusions from statements. It has three elements:

- **Syntax** — constructs forming a complete sentence (representation).
- **Semantics** — whether a sentence is meaningful and relates to the real world. 'Ram is riding a bike' is valid; 'Bike is riding Ram' is syntactically correct but semantically wrong.
- **Logical inference** — deducing conclusions using inference algorithms.

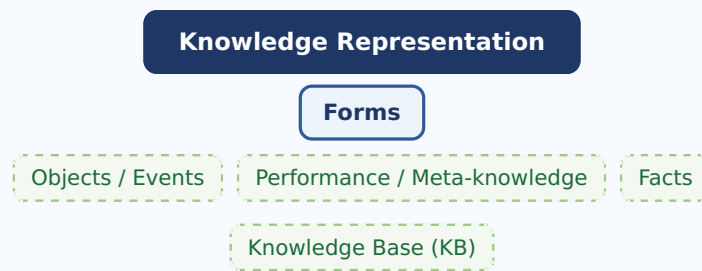
4.2.4 Cycle of Knowledge Representation in AI

An AI system has the components shown in Figure 4.3:



- **Perception** — gather sensor data, detect noise, define a response.
- **Learning** — improve from data; acquire knowledge and heuristics.
- **Knowledge Representation & Reasoning** — the core of human-like intelligence (forms in Figure 4.4).
- **Planning & Execution** — select states/actions and execute to reach the goal.

Figure 4.4 Forms of Knowledge Representation



4.2.5 Knowledge Representation Requirements

1. Representational accuracy — represent all required knowledge.
2. Inferential adequacy — produce new knowledge from existing structures.
3. Inferential efficiency — direct inference to appropriate results.
4. Acquisitional efficiency — easily acquire new knowledge automatically.

Bloom's Taxonomy Questions — 4.1 & 4.2 Knowledge Representation

- 1 **L1** Define knowledge representation and name the types of knowledge.
- 2 **L2** Explain why KR is more than database storage.
- 3 **L3** Identify objects, events, facts, meta-knowledge and KB for a self-driving car.
- 4 **L4** Differentiate syntax, semantics and logical inference with examples.
- 5 **L5** Evaluate representational accuracy and inferential adequacy.
- 6 **L6** Create a KR cycle for a book-recommendation AI.

GATE / Placement-Practice Questions — 4.1-4.2 KR Fundamentals

Each item is tagged with a Bloom level (L1-L6); the answer/hint follows in *italics*.

- 1 **L1** Knowledge about knowledge is called ____ knowledge. — *meta.*
- 2 **L1** 'How to do something' is: (A) declarative (B) procedural (C) structural (D) meta — *(B).*
- 3 **L2** Which KR property derives new knowledge? — *Inferential adequacy.*
- 4 **L3** 'Bike is riding Ram' is syntactically valid but semantically invalid because: — *it violates real-world meaning.*
- 5 **L4** Order the AI cycle: Learning, KRR, Perception, Planning. — *Perception → Learning → KRR → Planning & Execution.*
- 6 **L4** Match (1) rule-of-thumb (2) attributes (3) kind-of — (a) structural (b) declarative (c) heuristic — *1-c, 2-b, 3-a.*
- 7 **L5** Why is a database alone insufficient for an agent? — *It stores facts but lacks inference, learning and action.*
- 8 **L6** Design a KR cycle for a book recommender. — *Perceive borrowings/ratings; learn preferences; represent books/users; plan ranked recommendations; gather feedback.*

Previous Year VTU Questions — 4.1-4.2 KR

Curated from VTU B.E./B.Tech papers (AI / Machine Learning / AI & Applications). Marks and scheme indicated where known.

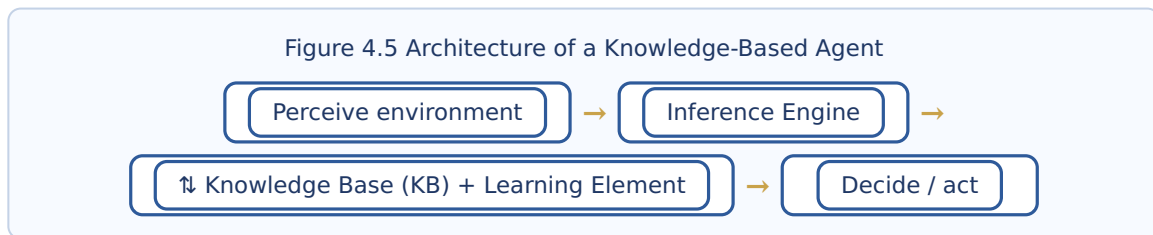
- What is KR? Explain what knowledge must be represented. (Jul 2024, 8 marks)
- Explain the types of knowledge with examples. (Jan 2024, 8 marks)
- What is logic? Explain syntax, semantics and inference. (Dec 2022, 8 marks)
- Explain the KR cycle in AI with a diagram. (Jun 2023, 10 marks)
- Discuss the properties of a good KR system. (Jun 2024, 6 marks)

4.3 Knowledge-Based Agent (KBA)

Knowledge-based agents mimic human knowledge, using it and reasoning to act efficiently. They maintain internal knowledge, reason with it, update it after observations, and take actions.

4.3.1 Architecture of a KBA

A KBA contains a knowledge base (KB) and an inference engine (IE) (Figure 4.5).



- **Knowledge base** — stores real-world facts as sentences in a KR language.
- **Inference engine** — derives new knowledge; the agent perceives, the IE consults the KB, and a learning element updates it. New sentences follow via forward/backward chaining.

4.3.2 Operations Performed by a KBA

- **TELL** — adds knowledge / records the selected action to the KB.
- **ASK** — queries the KB for the action to perform.
- **PERFORM** — executes the selected action.

4.3.3 A Generic KBA

The generic KB-AGENT (Figure 4.6) maintains a persistent KB and time counter t . For each percept it TELLS a percept sentence, ASKS for an action, TELLS the action sentence, increments t and returns the action.

```

function KB-AGENT(percept):
    persistent: KB, t = 0
    TELL(KB, MAKE-PERCEPT-SENTENCE(percept, t))
    action = ASK(KB, MAKE-ACTION-QUERY(t))
    TELL(KB, MAKE-ACTION-SENTENCE(action, t))
    t = t + 1
    return action

```

4.3.4 Levels of a KBA

Table 4.2 Levels of a Knowledge-Based Agent (Figure 4.7 illustrated by the AC example)

Level	Description	AC example
Knowledge	What the agent knows and its goals; fixes behaviour.	'It adjusts temperature per the weather.'
Logical	Knowledge encoded into logical sentences.	Deduces when/how to adjust.
Implementation	Physical execution of knowledge/logic.	Actual thermostat adjustment.

Figure 4.7 maps the three levels onto an automated air conditioner: the knowledge level states 'it adjusts temperature according to the weather'; the logical level deduces when and how to adjust; the implementation level physically changes the thermostat setting.

4.3.5 Approaches to Designing a KBA

- **Declarative** — start with an empty KB and TELL it sentences one by one.
- **Procedural** — encode desired behaviour directly as code (LISP, Prolog).
- **Hybrid** — practically used — declarative knowledge compiled into efficient procedural code.

Bloom's Taxonomy Questions — 4.3 Knowledge-Based Agent

- L1** List the components of a KBA.
- L2** Explain the roles of the KB and inference engine.
- L3** Use TELL, ASK and PERFORM to explain KBA response to a percept.
- L4** Analyse the knowledge, logical and implementation levels.
- L5** Evaluate the hybrid declarative+procedural approach.
- L6** Design a KBA for an automated AC or smart classroom.

GATE / Placement-Practice Questions — 4.3 KBA

Each item is tagged with a Bloom level (L1-L6); the answer/hint follows in *italics*.

- 1 **L1** The two core KBA components are ____ and _____. — knowledge base; inference engine.
- 2 **L1** Which operation adds facts? (A) ASK (B) TELL (C) PERFORM (D) INSERT — (B).
- 3 **L2** What does ASK return? — The action the agent should perform.
- 4 **L2** Forward/backward chaining are methods of the _____. — inference engine.
- 5 **L3** For a percept at time t , list the three steps. — TELL percept; ASK action; TELL action; return action, $t++$.
- 6 **L4** Distinguish the three levels for a medical KBA. — Knowledge=symptoms/rules; logical=encoded sentences; implementation=diagnosis code.
- 7 **L5** Why is a hybrid approach preferred? — Declarative gives flexible/explainable knowledge; procedural gives efficient execution.
- 8 **L6** Design a smart-classroom KBA. — KB: occupancy/schedule/temp; rules infer devices; PERFORM switches; update KB from sensors.

Previous Year VTU Questions — 4.3 KBA

Curated from VTU B.E./B.Tech papers (AI / Machine Learning / AI & Applications). Marks and scheme indicated where known.

- Explain KBA architecture with a diagram. (Jul 2024, 10 marks)
- What are TELL, ASK and PERFORM? Explain the generic KBA program. (Jan 2024, 8 marks)
- Explain the knowledge, logical and implementation levels with an example. (Dec 2022, 8 marks)
- Compare declarative, procedural and hybrid KBA approaches. (Jun 2023, 8 marks)

4.4 Types of Knowledge

Different types of knowledge can be expressed in a KR system.

Simple Relational Knowledge

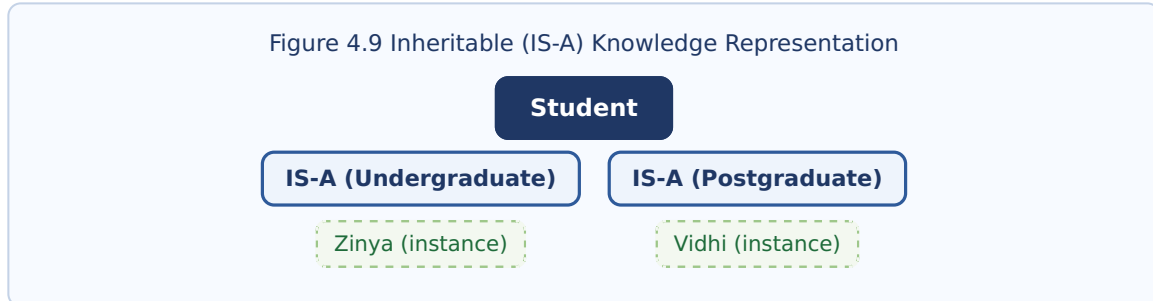
Facts about objects are stored as relations (tables) in databases (Figure 4.8); entity relationships are depicted, but inference is limited.

Figure 4.8 / Table 4.3 Simple Relational Knowledge — a students relation

USN	Name	Programme	CGPA
1KG21CS001	Zinya	B.E. CSE	8.9
1KG21CS014	Vidhi	B.E. CSE	9.2
1KG22AI007	Diya	B.Tech AI	8.5

Inheritable Knowledge

Inheritable knowledge stores data in a class hierarchy from general to specialised classes, showing the instance/IS-A relation (Figure 4.9); properties are inherited down the hierarchy.



Inferential Knowledge

Inferential knowledge uses formal logic to derive facts. Example: ‘Diya is a student’ and ‘All students are bright’ imply ‘Diya is bright’ (Student(Diya); $\forall x$ Student(x) $\rightarrow$ Bright(x); therefore Bright(Diya)).

Table 4.4 Relational, Inheritable and Inferential Knowledge

Type	Representation	Inference?	Example
Simple relational	Relational tables	Very little	Student table of names/ USN/marks
Inheritable	IS-A class hierarchy	Inherits properties down the hierarchy	UG-Student IS-A Student; Zinya IS-A UG-Student
Inferential	Formal logic	Derives new facts	Student(Diya); $\forall x$ Student(x) $\rightarrow$ Bright(x); $\vdash$ Bright(Diya)

Bloom’s Taxonomy Questions — 4.4 Types of Knowledge

- 1 **L1** List the types of knowledge used in KR systems.
- 2 **L2** Explain relational, inheritable and inferential knowledge.
- 3 **L3** Represent student details in a relational table.
- 4 **L4** Differentiate relational, inheritable and inferential knowledge.
- 5 **L5** Judge which representation best suits a class hierarchy and why.
- 6 **L6** Create an IS-A hierarchy for an academic institution.

GATE / Placement-Practice Questions — 4.4 Types of Knowledge

Each item is tagged with a Bloom level (L1-L6); the answer/hint follows in italics.

- 1 **L1** Tabular entity facts are called ____ knowledge. — simple relational.
- 2 **L2** IS-A hierarchies characterise ____ knowledge. — inheritable.
- 3 **L2** Formal logic that derives new facts is ____ knowledge. — inferential.
- 4 **L3** Given Student(Diya) and $\forall x \text{ Student}(x) \rightarrow \text{Bright}(x)$, infer: — Bright(Diya).
- 5 **L4** Strongest vs. least inference among the three? — Inferential strongest; simple relational least.
- 6 **L5** Why is inheritable knowledge suited to taxonomies? — Properties are inherited down IS-A links, avoiding duplication.
- 7 **L6** Build Person→Student→UG/PG with an inherited property. — e.g., Student inherits 'has roll number'; Zinya inherits it.

Previous Year VTU Questions — 4.4 Types of Knowledge

Curated from VTU B.E./B.Tech papers (AI / Machine Learning / AI & Applications). Marks and scheme indicated where known.

- Explain simple relational, inheritable and inferential knowledge with examples. (Jul 2024, 8 marks)
- What is inheritable knowledge? Represent a student IS-A hierarchy. (Jan 2024, 8 marks)
- Explain inferential knowledge with formal-logic examples. (Dec 2022, 6 marks)
- Compare relational, inheritable and inferential knowledge. (Jun 2023, 8 marks)

Topic Coverage & Quality Checklist

The table below audits every Module 1 topic against the five deliverables: formatted text, numbered tables/figures, Bloom's Taxonomy questions (L1-L6), GATE/ placement questions mapped to Bloom's, and previous-year VTU questions.

Table 5.1 Module 1 Topic-Coverage Checklist

Topic	Text	Tables/ Figures	Bloom L1-L6	GATE/ Placement	VTU PYQs
1.1 What is AI?	✓	T1.1	✓	✓	✓
1.1.1 How AI Works	✓	—	✓	✓	✓
1.1.2 Advantages/ Disadvantages	✓	T1.2	✓	✓	✓
1.2 History of AI	✓	T1.3 (text timeline)	✓	✓	✓
1.3 Types of AI	✓	Fig1.2 (tree), T1.4–1.5	✓	✓	✓
1.4 AI vs Augmented/ Cognitive & ML	✓	Fig1.4–1.6 (flows), T1.6– 1.7	✓	✓	✓
1.5 ML and Deep Learning	✓	Fig1.7 (flow)	✓	✓	✓
3.1 Defining Intelligence	✓	T3.1	✓	✓	✓
3.2 Components of Intelligence	✓	Fig3.1 (flow), T3.2	✓	✓	✓
3.3 Human vs Machine Intelligence	✓	T3.3	✓	✓	✓
3.4 Agent & Environment	✓	Fig3.2–3.8 (text diagrams), T3.4	✓	✓	✓
3.4.5–3.4.6 Environments	✓	Fig3.9–3.12 (text), T3.5	✓	✓	✓
3.5 Search & Properties	✓	Fig3.13 (tree)	✓	✓	✓
3.6 Uninformed Search	✓	Fig3.14–3.18 (tree figures), T3.6	✓	✓	✓
3.7 Informed Search	✓	—	✓	✓	✓

Topic	Text	Tables/ Figures	Bloom L1-L6	GATE/ Placement	VTU PYQs
4.1 Knowledge & Intelligence	✓	Fig4.1 (flow)	✓	✓	✓
4.2 Knowledge Representation	✓	Fig4.2-4.4 (text diagrams), T4.1	✓	✓	✓
4.3 Knowledge-Based Agent	✓	Fig4.5 (flow), T4.2	✓	✓	✓
4.4 Types of Knowledge	✓	Fig4.8-4.9 (table & tree), T4.4	✓	✓	✓

Summary:

- All 19 topic sections of Module 1 (Parts A, B, C) are covered with formatted, selectable text (text-heavy source images were converted to real text, tables and diagrams).
- Numbered tables and figures are included throughout; node-and-edge search trees (DFS, DLS, BFS, UCS) are retained as figures, while box/timeline diagrams are redrawn as text diagrams.
- Every topic carries Bloom's questions across L1-L6, now as compact bullet lists.
- GATE/placement questions (tagged with Bloom levels, with answer hints) and previous-year VTU questions accompany every topic in the same compact list format.

GATE / Placement Preparation Pointers

- Search algorithms (3.6-3.7) are highest-yield for GATE: master Table 3.6 — data structure, completeness, optimality, $O(b^d)$, $O(bm)$, $O(b^{(d/2)})$.
- Practise numericals: BFS/DFS orders, UCS least-cost paths, greedy node selection, IDDFS iterations, bidirectional meeting points.
- Agents (3.4): PEAS descriptions and environment classification are common 1-2 mark questions.
- Knowledge representation (Part C): types of knowledge, KBA architecture, TELL/ASK/PERFORM and logical inference.
- Use L5/L6 questions for interviews — they test design reasoning (PEAS for a taxi, ML problem formulation).

References & Source

- Lokesh M R (2026). Introduction to AI and Applications (Module 1) — Beyond Classical AI. First Edition. VTU Textbook Series, CSE. Zenodo. <https://doi.org/10.5281/zenodo.20391293>
- Visvesvaraya Technological University, Belagavi — B.E./B.Tech syllabus, Course Code 1BAIA103/203 (2025-2026 scheme).
- VTU previous-question papers for AI / Machine Learning / AI & Applications (2022 & 2018 schemes).
- Russell, S. & Norvig, P., Artificial Intelligence: A Modern Approach (agent, search and KR concepts referenced in GATE questions).

Disclaimer: GATE and VTU questions in this edition are practice/illustrative items mapped to the syllabus; VTU wording and marks are indicative and should be verified against the official VTU question-paper bank.